OPENCAMPUS.SH · SESSION 1

Introduction to Data Science and Machine Learning

What data science is for, your client, and your first question

TODAY

Plan for the session

1. Welcome: who we are and how the course works
2. Getting to know each other: what is AI?
3. The project and the client
4. Example notebook: pedestrians in Kiel’s Holstenstraße
5. Your questions and the canvas
6. Live demo: working with a coding agent
7. Homework

WELCOME

Your instructors

Giulia Sepeda

Data scientist: analytics, predictive modelling and time series forecasting, with a clinical background

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Steffen Brandt

Educational research, data science and AI, at opencampus.sh

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And you? Write in the chat where you are joining from.



WHO WE ARE

We are opencampus.sh

A non-profit organisation in Kiel that runs a variety of initiatives.

Education, support and networking for entrepreneurs, creatives and anyone curious, whatever their age, education or origin.

Open to everyone, and mostly free.

The goal: a stronger entrepreneurial landscape, creative change, and an innovative, sustainable future.

OPENCAMPUS.SH

Places to learn, work and make



starterkitchen
event and meeting rooms



COBL
cozy working, culture and events



Fablab.sh
3D printers and tools

OPENCAMPUS.SH · KIEL, BY THE WATER

Waterkant Festival

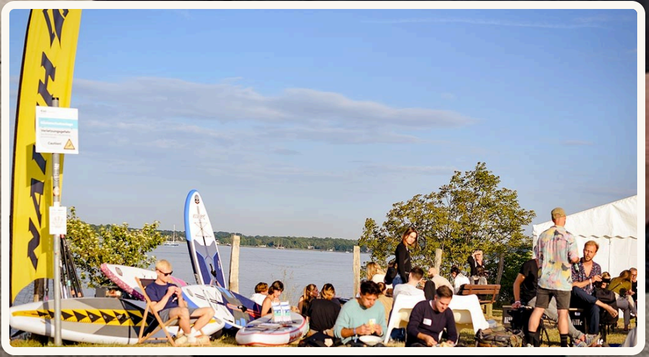
Two days of talks, pitches and hands-on experiences: the meeting point of the start-up and innovation scene in Schleswig-Holstein.

2026: 1,600+ visitors · 230+ speakers · 90+ startups

Dates for the next edition will follow · waterkant.sh







MACHINE LEARNING AT THE WATER'S EDGE

Coding.Waterkant

A four-day coding week: build AI projects with others, join hands-on workshops, share meals and present what you built.

A community of AI practitioners since 2019. Educational leave possible.

Dates for the next edition will follow · coding.waterkant.sh







“Education is the only thing people try to get less out of than they pay for.”

Get the most out of this course: ask questions, and answer them.

WELCOME

What you will be able to do

Code with an agent

Python analyses with a coding agent, and checks that show the results are right.

Work as a team

Git and GitHub: what lives on your laptop, what lives on GitHub.

Describe data

Charts and statistical tests that answer a question.

Prepare data

Encoding, feature engineering and missing values.

Build models

From a baseline and linear regression to a neural network.

Ask and tell

Turn a decision into questions, know what the data cannot answer, and explain the result.

FORMAT

A flipped classroom

Before

Watch the videos and do the readings.

Post your questions in the Element course room the day before.

In the session

A 10-minute video check: a short quiz and your posted questions.

A short input, then mostly group work with the agent and sharing results.

After

Finish the tasks of the week.

Prepare a short answer to the question of the week; we discuss it in class.

Sessions are for applying what you learned, not for repeating the videos.

ORGANISATION

Where things live

Tool	What it is for
Course handbook	One page per week: tasks, videos, links and the slides
Element	Course room for all questions, and one room per team
EduHub	Enrolment, attendance and project submission
DeepLearning.AI	The learning videos (free account)
GitHub	Your team repository, from week 3
Kaggle	The course competition for the sales forecast

Course handbook: opencampus.gitbook.io → Introduction to Data Science and ML

ORGANISATION

What you need for the certificate

1 · Attend at least 80 % of the sessions (miss at most two).

2 · Complete the weekly tasks.

3 · Present your team project in session 10; every member presents a part.

4 · Submit the project on EduHub, with a public GitHub repository that every member has contributed to.

5 ECTS

about 150 hours in total, about 6.5 hours this week

GROUP WORK · 15 MINUTES · GROUPS OF 3–4

What is AI?

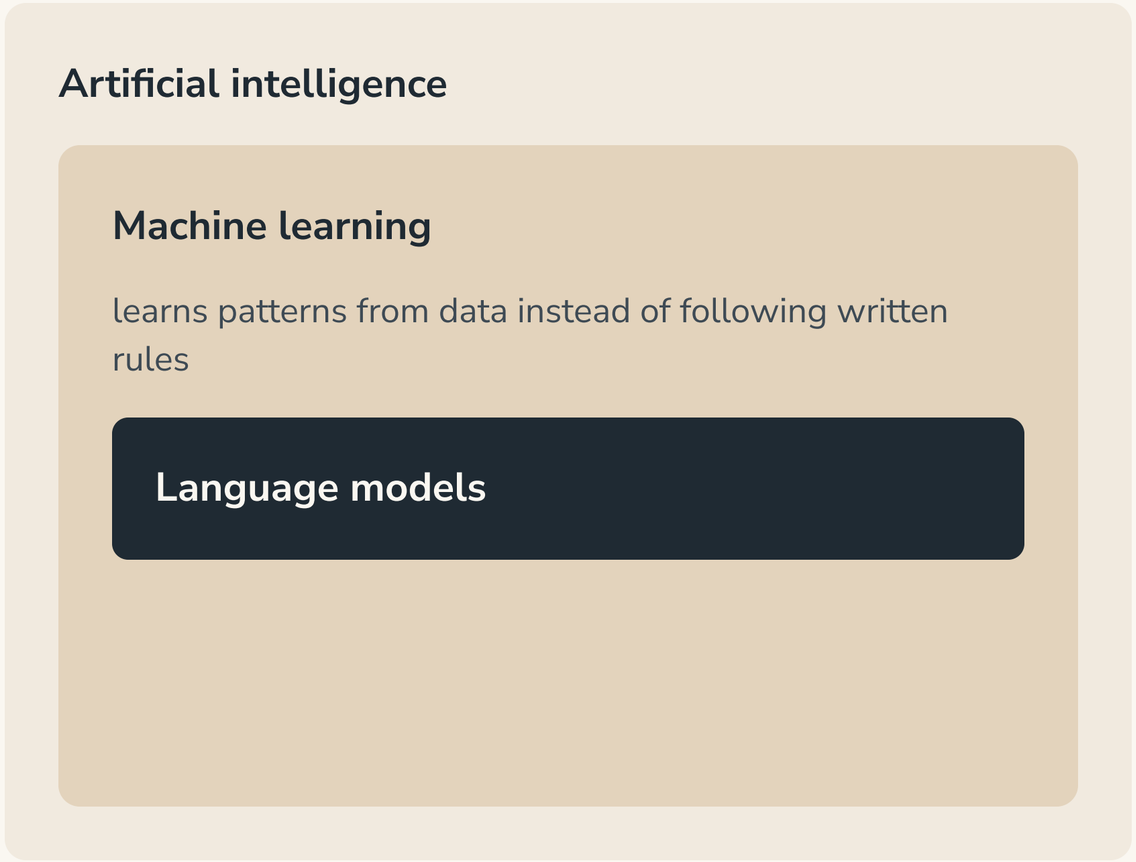
1 · Introduce yourselves: your background and why you are here.

2 · Agree on a one-sentence definition and one everyday example each for **AI**, **machine learning**, **data science** and **AI agent**.

3 · Post the examples in the Element course room.

PLENARY · 5 MINUTES

How the terms relate



An AI agent is



The harness gives the model tools, such as running code or editing files, and runs the loop: plan, act, look at the result, adjust.

The agent acts in an **environment**: your project folder and terminal, or a sandbox in the cloud.

Example: Claude Code is the harness, Claude Opus 5.5 the model.

Data science uses statistics and machine learning to answer questions and support decisions.

WHY DATA SCIENCE

Data is worth something when someone acts on it

Decision	→	Question	→	Data	→	Analysis	→	Advice
What will someone do differently?		What do they need to know?		What can it show, and what not?		Checked, not just run		Clear, with its limits
How much bread to bake for tomorrow		*Expected sales tomorrow, per product*		*Sales and weather: yes. Prices and special offers: no*		*A forecast that beats “same as last week”*		*How much to bake, and when not to trust it*

In this course, every task starts from a decision, not from a method.

THE PROJECT

A sales forecast for a bakery in Kiel

6
product groups

5 years
of daily turnover, 2013–2018

+
weather and Kiel Week dates

Bread · Rolls · Croissants · Confectionery · Cakes · Seasonal bread

In teams of three or four, from week 3, you forecast tomorrow’s turnover for each product group, and compete on Kaggle with your forecasts for the following year.

THE PROJECT TASK

What your team delivers by week 10

Forecast

Next-day turnover for each of the six product groups. On Kaggle (MAPE): a whole year ahead, without recent sales.

Presentation

8–10 minutes for the operations manager in session 10. Every team member presents a part.

Repository

Public on GitHub, with documented code and READMEs a data team could take over.

Submission

On EduHub: documentation, presentation, repository link and cover image, sent by one member for the team.

MILESTONES

Week 3 teams

Week 5 baseline

Weeks 6–7 linear model

Week 8 neural network

Week 9 missing values

Week 10 presentation

YOUR CLIENT

The operations manager of a bakery in Kiel

Every afternoon, the operations manager decides how much of each product group is baked for the next day.

Too much

Waste

Too little

Lost sales and disappointed customers

What the manager needs from you

Clear recommendations, a sense of how reliable they are, and no jargon: the manager is not a data scientist.

WORKING WITH AN AGENT

When the agent writes the code, what is left for you?

The agent

Writes and runs the code

Looks up facts in the data

Drafts and recommends

You

Ask a question worth answering

Make the decisions

Check that the results are right, and explain them

Code that runs without errors can still be wrong. Agreeing on checks is your job.

5 MINUTES

Break

GROUP WORK · 25 MINUTES · SAME GROUPS

Pedestrians in Kiel’s Holstenstraße

1 · Share: each of you presents the pattern and the question you noted in the preparation.

2 · Explain: for each pattern, what could cause it? Check with the agent where you are unsure.

3 · Sort: which of your questions can this data answer, and which need more data? Pick your best question for the plenary.

4 · Optional: let the agent add one plot that answers one of your questions.

PLENARY · 10 MINUTES

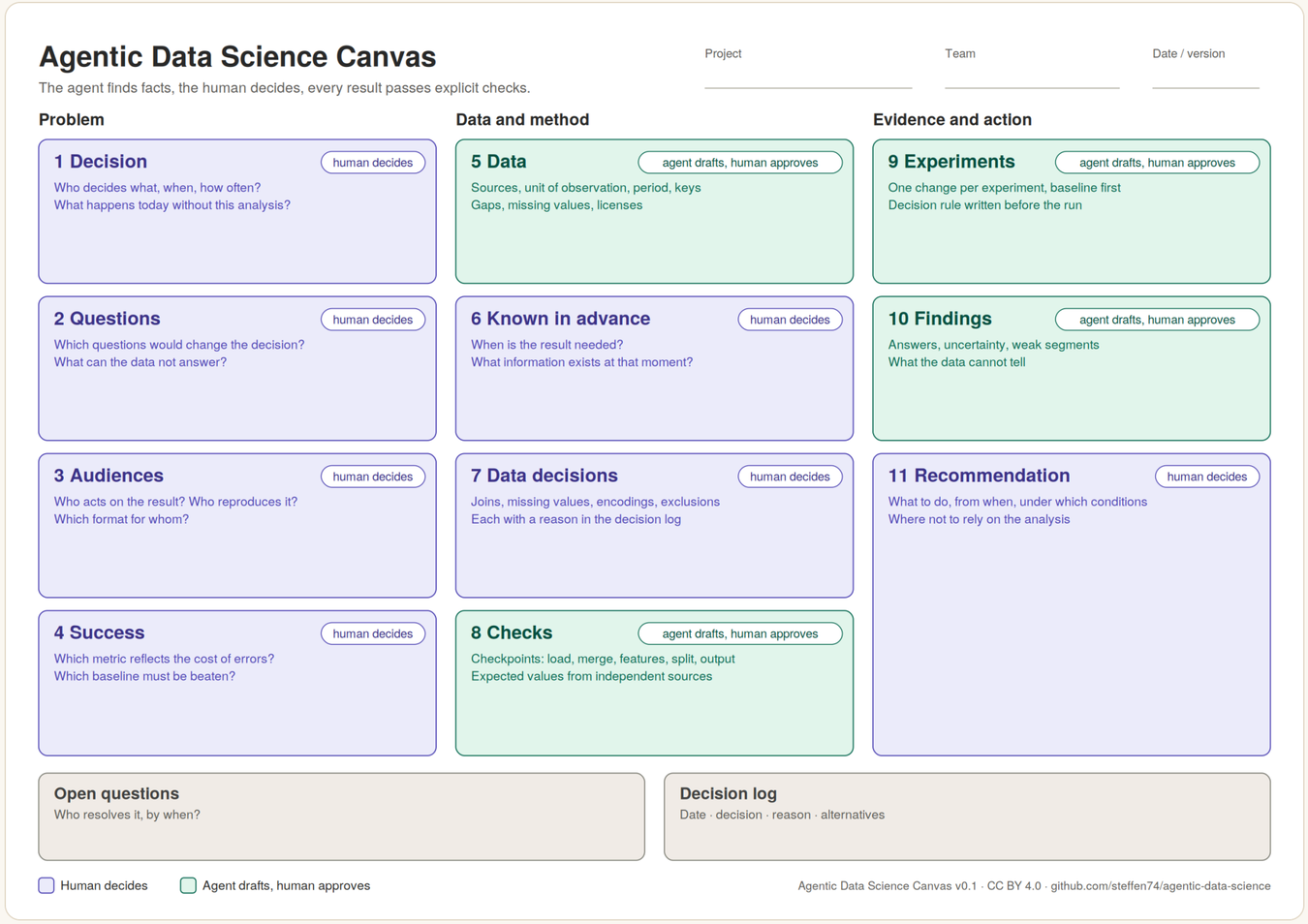
Your best question

?

The question a bakery owner in the street would ask

+

What you would need to answer it, beyond this data



YOUR PROJECT, ON ONE PAGE

The Agentic Data Science Canvas

Problem: decision, questions, audiences, success

Data and method: data, what is known in advance, data decisions, checks

Evidence and action: experiments, findings, recommendation

Open it on your laptop: github.com/steffen74/agentic-data-science → canvas (PDF)

THE CANVAS

Every block has an owner

You decide

Decision · Questions · Audiences · Success · Known in advance · Data decisions · Recommendation

The agent may recommend, then waits for your answer.

The agent drafts, you approve

Data · Checks · Experiments · Findings

Nothing counts until you have checked it.

This week: blocks 1–4, the problem.

LIVE DEMO · 10 MINUTES

Sharpening a question with an agent

```
/grill-question How much of each product group should we bake for tomorrow?
```

1 · The agent asks in rounds, each question with its recommended answer.

2 · We answer, and overrule one recommendation.

3 · CANVAS.md fills up, block by block.

4 · Then, on the seasonal bread question, the agent analyses: plan, implement, run, verify.

YOUR PROJECT QUESTION · FOR YOUR TEAM, FROM WEEK 3

“How much of each product group should we bake for tomorrow?”

QUESTION OF THE WEEK · FOR YOUR PRACTICE CANVAS AND THE DATA

“Should we start selling our seasonal bread in October, or only in November?”

Seasonal bread is product group 6 in umsatzdaten_gekuerzt.csv.

HOMEWORK · ABOUT 3 HOURS

Until next session

1 · Set up

Clone the template repository next to your course resources folder, load the skills as described on the week 1 page, then run /setup-ds-project.

2 · Sharpen

Run /grill-question with the question of the week typed after the command, until blocks 1–4 of CANVAS.md are filled. Disagree with at least one recommendation, and say why.

3 · Analyse

Watch the three t-test videos first. Question of the week: when was the seasonal bread sold in past years? Compare average daily sales in October and November, and test whether they differ significantly.

4 · Check

Let the agent explain each step. Are the results plausible? What can the data tell the manager, and what not?

DELIVERABLE · FOR THE OPERATIONS MANAGER

Three or four sentences

A clear recommendation

October or November, and why.

One supporting number

The one that matters most for the decision.

Written for the manager

No jargon: the manager is not a data scientist.

What the data cannot tell

One limit the manager should know.

Write it in the last cell of your notebook and bring it to the next session.

BEFORE NEXT SESSION · T-TEST VIDEOS BEFORE TASK 3

Videos for this week

Course	Lessons	Min
Claude Code course	Adding Features · Testing, Error Debugging and Code Refactoring	29
Python for Data Analytics, module 2	Beyond lists · Importing modules · Pandas · Reading CSV into Python · DataFrames · Attributes and methods · Selecting columns · Counts, sums and histograms	25
Python for Data Analytics, module 4	Confidence intervals · One-sample t-tests · Two-sample t-tests	15

Plan about 1.7 hours with pausing and following along. All videos are on DeepLearning.AI with a free account; the links are on the course page for week 1.

LAST 5 MINUTES

Questions

Setup not working? Stay now, and we look at it together.

Questions about the videos? Post them in the Element course room by the day before the session.

Next week: data import and visualization.